

Praveen Gadiparthi



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Raleigh, NC

PROFESSIONAL SUMMARY:

- Cloud & DevOps Engineer with 6+ years of experience designing, deploying, automating and operating secure, highly available, scalable, and cost-optimized cloud architectures across AWS and Azure environments.
- Designed and implemented multi-account AWS Landing Zones using AWS Organizations, SCPs, centralized logging, and governance best practices, along with Azure Landing Zones using Management Groups, Azure Policy, and centralized monitoring frameworks.
- Architected secure hybrid networking solutions using Amazon VPC, subnets, NAT/Internet Gateways, Transit Gateway, Private Link, Site-to-Site VPN, and Direct Connect, as well as Azure VNets, NSGs, Azure Firewall, Private Endpoints, VPN Gateway, and ExpressRoute.
- Implemented least-privilege identity and access management models using AWS IAM roles and policies, IAM Identity Center (SSO), and Azure AD (Entra ID), Managed Identities, RBAC, and Conditional Access policies.
- Optimized cloud cost and performance using AWS Cost Explorer, Auto Scaling, Reserved/Spot Instances, and Azure Cost Management, VM scaling sets, and reserved capacity planning.
- Designed and automated disaster recovery and business continuity strategies using AWS Backup, cross-region replication, snapshot lifecycle policies, and Azure Site Recovery, geo-redundant storage (GRS), and automated failover mechanisms.
- Provisioned and managed infrastructure using Terraform with reusable modular architecture, remote state management (S3/DynamoDB and Azure Storage), and environment segregation across Dev/QA/UAT/Prod.
- Implemented Terraform Enterprise/Cloud governance with RBAC controls, Sentinel policy-as-code enforcement, and CI/CD integration for AWS and Azure infrastructure deployments.
- Designed and maintained CI/CD pipelines using AWS CodePipeline, Jenkins, GitHub Actions, and Azure DevOps for automated build, test, infrastructure provisioning, and application deployments.
- Integrated Maven, Ant, and SonarQube into pipelines to enforce automated builds, code quality gates, vulnerability scanning, and secure artifact management.
- Built serverless and event-driven architectures using AWS Lambda, API Gateway, SQS/SNS, Event Bridge, along with Azure Functions, Event Grid, Service Bus, and Logic Apps.
- Managed highly available relational databases using Amazon RDS (MySQL, PostgreSQL, Aurora) and Azure SQL, PostgreSQL Flexible Server, with Multi-AZ/Zone redundancy, read replicas, automated backups, and performance tuning.
- Implemented Blue/Green and Canary deployment strategies using Code Deploy, Kubernetes rolling updates, Route53 weighted routing, Azure Traffic Manager, and AKS deployment strategies to achieve zero-downtime releases.
- Supported application modernization initiatives by migrating monolithic applications to microservices and container-based architectures using EKS, AKS, and cloud-native design principles.
- Implemented and governed GenAI workloads using Amazon Bedrock and Azure OpenAI, enforcing secure IAM/RBAC-based access controls, private networking, and responsible AI guardrails in enterprise environments.
- Developed automation scripts, Lambda functions, and Azure Functions using Python (Boto3/Azure SDK) for infrastructure orchestration, compliance validation, cost reporting, and operational automation.
- Implemented centralized monitoring and observability using CloudWatch, CloudTrail, AWS Config, Azure Monitor, Log Analytics, Prometheus, Grafana, and distributed tracing for full-stack visibility.
- Led production incident response efforts across AWS and Azure, conducted root cause analysis (RCA), improved MTTR, automated remediation workflows, and created operational runbooks to enhance system reliability and release stability.

TECHNICAL SKILLS:

Cloud Platforms: AWS (EC2, S3, VPC, IAM, Lambda, EKS, ECS, CloudWatch, CloudTrail, SQS, SNS, API Gateway, CodePipeline), Azure (VMs, AKS, Azure Functions, App Service, Storage, Azure SQL, Service Bus, Azure Monitor)

CI/CD & DevOps: AWS CodePipeline, Code Build, Code Deploy, Azure DevOps Pipelines, GitHub Actions, Jenkins, YAML pipelines, Blue-Green and Canary deployments

Infrastructure as Code: Terraform, AWS CloudFormation, Azure ARM Templates, Bicep, reusable modules, multi-environment provisioning

Containers & Orchestration: Docker, Kubernetes, Amazon EKS, Azure AKS, ECS, Helm charts, ECR, ACR

Monitoring & Logging: Amazon CloudWatch, AWS CloudTrail, Azure Monitor, Log Analytics, Application Insights, Prometheus, Grafana

Security & Identity: AWS IAM (roles, policies, SSO), Azure AD (Entra ID), Managed Identities, Secrets Manager, SSM Parameter Store, Azure Key Vault

Networking: AWS VPC, Transit Gateway, VPC Peering, Direct Connect, Azure VNet, ExpressRoute, Private Endpoints

Automation & Scripting: Python, Bash, PowerShell, Shell scripting.

Reliability & DR: AWS Backup, snapshot automation, cross-region replication, Azure Site Recovery, HA/DR design

WORK EXPERIENCE:

**Role: Cloud & DevOps Engineer
(Contract | Remote)**

**Client: State Of Illinois
(DCFS)**

Raleigh, NC, USA 08/2024 – Present

- Designed and configured Amazon VPCs with public/private subnets, route tables, NAT/Internet Gateways, and Site-to-Site VPN connections for secure hybrid cloud connectivity.
- Implemented IAM least-privilege policies, Security Groups, NACLs, KMS, AWS WAF, and Secrets Manager to enforce cloud security best practices.
- Managed highly available EC2, RDS, and S3 resources across multiple Availability Zones with automated backup, snapshot, and cross-region replication strategies.
- Built reusable Terraform modules for VPCs, EKS clusters, IAM roles, RDS, S3, and Secrets Manager to standardize deployments across Dev/QA/UAT/Prod environments.
- Automated infrastructure provisioning using Terraform integrated with GitHub Actions and AWS CloudFormation.
- Applied Terraform Enterprise/Cloud governance with RBAC, workspace policies, and policy-as-code enforcement.
- Developed CI/CD pipelines using GitHub Actions for automated build, test, and deployment workflows.
- Implemented Git branching strategies with Pull Requests, code reviews, and pipeline-based quality gates using Maven, Ant, and SonarQube.
- Automated containerized application deployments using Docker images pushed through GitHub Actions to AWS EKS environments.
- Deployed and managed containerized workloads using Docker, EKS, Kubernetes, and Helm charts with auto-scaling, rolling updates, and zero-downtime releases.
- Configured Kubernetes Network Policies and Pod Security Standards and admission controllers to enforce workload isolation and least privilege access.
- Built serverless architectures using AWS Lambda, API Gateway, SQS, SNS, and Event Bridge for event-driven microservices.
- Managed highly available Amazon RDS (MySQL, PostgreSQL, Aurora) with Multi-AZ deployments, read replicas, automated backups, and performance tuning.
- Implemented Blue/Green and Canary deployments using Kubernetes rolling updates and Route53 weighted routing to reduce downtime and deployment risk.
- Supported deployment of GenAI/ML pipelines using Amazon SageMaker, Bedrock, and Lambda, integrating model training, fine-tuning, and endpoint provisioning with CI/CD.
- Developed Python scripts and AWS Lambda functions for automation of infrastructure provisioning, IAM audits, container scanning, and compliance checks.
- Built Python utilities to integrate security scanning (Trivy), Secrets Manager, and CI/CD pipelines for operational efficiency.
- Configured CloudWatch, Prometheus, Grafana, X-Ray, OpenSearch (ELK), and Splunk for centralized monitoring, logging, and alerting of AWS workloads.
- Participated in production incident response, performing root cause analysis (RCA), implementing fixes to reduce MTTR, and creating operational runbooks.

Environment (AWS):

AWS (EC2, VPC, IAM, EKS, ECS, Lambda, RDS, S3, CloudWatch, CloudTrail, WAF, Secrets Manager, KMS, CodePipeline, Code Build, Code Deploy, Event Bridge, SageMaker), Git, GitHub, Jenkins, GitHub Actions, Ansible, Docker, Kubernetes, Helm, Terraform, CloudFormation, Prometheus, Grafana, Splunk, Dynatrace, Maven, Groovy, Shell, Python.

**Role: DevOps Engineer
(Contract | Hybrid)**

Client: Comcast

Philadelphia, PA, USA 07/2023 – 08/2024

- Managed and resolved cloud infrastructure requests by attending to AWS and Azure service tickets, ensuring timely and accurate issue resolution.
- Designed and configured secure networking solutions using AWS VPC, Azure VNets, subnets, route tables, NAT/Internet Gateways, NSGs, Azure Firewall, and Site-to-Site VPN for hybrid cloud connectivity.
- Built reusable Terraform modules to provision standardized VPCs, VNets, EKS/AKS clusters, IAM/Role Assignments, RDS/Azure SQL, and Secrets/Key Vault across Dev, QA, UAT, and Prod environments.
- Integrated Terraform Enterprise/Cloud for workspace governance, RBAC, policy-as-code enforcement, and automated environment provisioning.
- Automated containerized deployments using GitHub Actions and Jenkins, deploying Docker workloads across AWS ECS/EKS and Azure AKS/App Services with Helm charts for multi-environment customization.
- Developed CI/CD pipelines integrating GitHub, Azure DevOps, Maven, Ant, and SonarQube to ensure automated build, test, quality, and deployment workflows.

- Deployed and managed Kubernetes clusters on EKS and AKS, implementing auto-scaling, rolling updates, Blue/Green and Canary deployment strategies, Pod Security Standards and admission controllers, and Network Policies for workload isolation.
- Built serverless and event-driven architectures using AWS Lambda, Azure Functions, API Gateway, Event Bridge, SQS, SNS, Glue, Data Factory, and Athena for ETL workloads, microservices, and analytical querying.
- Managed relational databases including Amazon RDS (MySQL, PostgreSQL, Aurora) and Azure SQL with Multi-AZ deployments, read replicas, automated backups, and performance tuning.
- Supported GenAI/ML pipelines on AWS SageMaker and Azure Machine Learning, automating model training, fine-tuning, endpoint provisioning, and continuous retraining using Terraform and CI/CD workflows.
- Developed Python-based AWS Lambda and Azure Functions to automate VM operations, identity audits, model retraining triggers, security scanning, and compliance validation.
- Implemented centralized logging, monitoring, and observability using CloudWatch, Azure Monitor, Log Analytics, Prometheus, Grafana and Splunk.
- Configured proactive alerts, dashboards, and anomaly detection for applications, containers, and infrastructure to improve operational visibility.
- Integrated container image scanning using Trivy with ECR/ACR, automating security checks within CI/CD pipelines.
- Automated server provisioning and configuration using Ansible, creating reusable roles, playbooks, and dynamic inventories across AWS and Azure VM environments.
- Developed reusable Jenkins pipelines and Groovy scripts to automate build, test, and deploy workflows, improving CI/CD efficiency and traceability.
- Participated in production incident response and troubleshooting, leveraging logs, metrics, and monitoring tools to restore services quickly and reduce MTTR.
- Collaborated with cross-functional teams to define DevOps roadmaps, standardize deployment patterns across AWS and Azure, and meet SLA/SLO objectives.
- Optimized cloud performance and cost using tagging strategies, right-sizing EC2/Azure VMs, Auto Scaling, Spot/Reserved Instances, and cost analysis tools.
- Conducted root cause analysis (RCA), created operational runbooks, and implemented preventive actions to enhance system reliability and DevOps best practices.

Environment (AWS + Azure):

Azure (VMs, VNETs, AKS, App Service, Azure Functions, Azure Storage, Azure SQL, Service Bus, Key Vault, Data Factory, Monitor, Log Analytics, Application Gateway, ACR, Azure ML),

Git, GitHub, Jenkins, GitHub Actions, Ansible, Docker, Kubernetes, Helm, Terraform, CloudFormation, ARM Templates, Prometheus, Grafana, Splunk, Dynatrace, Maven, Groovy, Shell, Python

AWS (EC2, VPC, IAM, EKS, ECS, Lambda, RDS, S3, CloudWatch, CloudTrail, WAF, Secrets Manager, KMS, CodePipeline, Code Build, Code Deploy, Event Bridge, SageMaker),

Role: Student Assistant (Part Time)

University of North Texas

Denton, TX, USA 01/2022 – 05/2023

- Managed and resolved cloud infrastructure requests by attending AWS service tickets, ensuring timely and accurate issue resolution across EC2, S3, RDS, Lambda, and VPC resources.
- Assigned, tracked, and managed AWS cloud resources, including IAM roles, S3 buckets, EC2 instances, RDS databases, and Secrets Manager entries, ensuring compliance with organizational policies.
- Designed and developed self-service AWS workflows using Lambda, API Gateway, and Step Functions to automate resource provisioning, approvals, and request tracking for business units.
- Maintained accurate cloud asset inventory across AWS environments, including EC2 instances, RDS/Aurora databases, S3 storage, EKS clusters, and container workloads.
- Collaborated with DevOps and cloud teams to streamline workflow automation, ticket routing, and escalations using AWS Lambda, Event Bridge, and CloudWatch alarms.
- Monitored SLA compliance for AWS cloud requests, leveraging CloudWatch metrics, alarms, and dashboards to ensure timely resolution and customer satisfaction.
- Automated repetitive cloud operations using Python scripts and Lambda functions, reducing manual effort in resource provisioning, monitoring, and compliance checks.
- Integrated Terraform and CloudFormation templates for standardized, repeatable deployments of VPCs, EC2 instances, RDS, and EKS clusters.

**Role: Cloud Engineer
(Full-Time | Onsite)**

Amara Raja Batteries Ltd

Tirupati, India 07/2019 – 11/2021

- Designed and implemented an Industrial IoT data pipeline to collect real-time telemetry (temperature, vibration, pressure) from factory equipment such as bearings, silos, and conveyor systems using edge sensors and IoT gateways.
- Configured Microsoft Azure IoT Hub to securely ingest high-volume telemetry from distributed factory devices and enable reliable device-to-cloud communication.
- Built Infrastructure as Code (IaC) using Terraform to provision Azure resources including Virtual Networks, Storage Accounts, Compute Instances, and AKS clusters across Dev, QA, and Production environments.
- Developed real-time data ingestion pipelines using Azure Event Hub and Azure Stream Analytics to process streaming device data and detect abnormal machine conditions.
- Stored large volumes of machine telemetry and historical operational data in Azure Blob Storage and Azure Data Lake Storage Gen2 for analytics and predictive maintenance workloads.
- Built containerized microservices using Docker and deployed them on Azure Kubernetes Service (AKS) to process streaming data and run anomaly detection services.
- Implemented machine condition monitoring algorithms to detect overheating, abnormal vibration patterns, and pressure spikes using historical telemetry trends.
- Developed Python-based data processing services to clean, transform, and analyze telemetry data and generate predictive maintenance alerts.
- Created CI/CD pipelines using GitHub Actions to automate infrastructure provisioning, container image builds, testing, and deployment of IoT data processing services.
- Implemented edge data buffering and batch synchronization mechanisms to handle intermittent connectivity from remote industrial devices.
- Configured observability and monitoring dashboards using Grafana and Azure Monitor to visualize real-time telemetry, infrastructure health, and anomaly alerts.
- Integrated container security scanning using Trivy within CI/CD pipelines to ensure secure container images before deployment to Kubernetes clusters.
- Designed secure networking architecture using Azure VNets, Network Security Groups (NSGs), and Private Endpoints to protect IoT data pipelines and cloud workloads.
- Implemented event-driven alerting workflows using Azure Functions and Azure Logic Apps to notify operations teams when sensor thresholds exceed safe limits.
- Built centralized logging and monitoring using Azure Log Analytics to track device connectivity, telemetry ingestion performance, and system reliability.
- Collaborated with mechanical and operations teams to integrate industrial sensor data with cloud analytics platforms to enable predictive maintenance and operational optimization.

Environment:

Azure (Virtual Machines, Virtual Networks, Network Security Groups, Azure IoT Hub, Event Hub, Stream Analytics, Azure Functions, Logic Apps, Data Factory, Azure Monitor, Application Insights, Azure Security Center, Azure Key Vault, Azure Kubernetes Service (AKS), Azure SQL Database, Service Bus, Event Grid, Blob Storage, Archive Storage, Azure Data Lake Storage), Terraform, Ansible, Docker, Kubernetes, Helm, Git, GitHub Actions, Grafana, Splunk, Python, Bash, Linux, Jira.

EDUCATION:

Master of Science

University of North Texas

Denton, TX, USA 05/2023

Major in Information Systems and Technologies

Bachelor of Science

**Jawaharlal Nehru
Technological University**

Anantapur, AP, INDIA 05/2019

Major in Mechanical Engineering

CERTIFICATIONS:

Certification Name: [AWS Certified Solutions Architect – Associate](#)

Certification Name: [Certified Kubernetes Administrator](#)

Certification Name: [HashiCorp Certified: Terraform Associate \(004\) \(HCTA0-004\)](#)

Certification Name: [Certified SAFe® 6 Practitioner](#)